



Sri Lanka Institute of Information Technology

M.Sc. in
Artificial Intelligence

Final Examination
Year 1, Semester 1 (2025)

IT5012 – Mathematics for Artificial Intelligence

Duration: 2 Hours

November 2025

Instructions to Candidates:

- ◆ This paper has Four questions.
- ◆ Answer all questions in the booklet given.
- ◆ The total mark for the paper is 40.
- ◆ This paper contains Four pages, including the cover page.
- ◆ Electronic devices capable of storing or retrieving text, including mobile phones, smart watches, and tablets, are not allowed. Calculators are allowed.
- ◆ A4-size reference sheet is allowed.

Question 01: Linear Algebra**(10 Marks)**

AI-based face recognition systems learn compact feature representations to identify individuals based on visual characteristics. In this question, each image is represented by a two-dimensional feature vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, where x_1 denotes the average intensity around the eyes and x_2 denotes the average curvature of the mouth. The model optimizes its representation by applying a linear transformation $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$, to enhance useful feature directions.

- (a) Show that T is a linear transformation and explain what linearity implies in this AI context, particularly in combining or scaling facial features. (2 marks)
- (b) To examine how the transformation reshapes the feature space:
 - i) Find the eigenvalues of A . (1 marks)
 - ii) Determine the corresponding normalized eigenvectors. (1 marks)
 - iii) Hence express A using its eigendecomposition $A = PDP^{-1}$. (2 marks)
 - iv) Discuss how eigenvalue-eigenvector pair represents a dominant direction of variation, indicating how certain feature combinations are amplified or suppressed. (1 marks)
- (c) Write $T(\mathbf{x})$ explicitly in terms of x_1, x_2 , then evaluate $T(\mathbf{x})$ at $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Finally, using the eigenvalues you found in (b), state which feature combination is emphasized by A . (3 marks)

Question 02: Calculus**(10 Marks)**

Gradient descent is one of the most common optimization techniques used in AI for training supervised learning models. In this question, you will train a simple linear model $\hat{y} = \theta x$ on a single training example $(x, y) = (2, 5)$. The model optimizes the parameter θ using gradient descent to minimize the squared-error loss function $L(\theta) = (\theta x - y)^2$.

- (a) State the definition of the derivative (as a limit) and explain briefly why $L(\theta)$ is continuous and differentiable for all $\theta \in \mathbb{R}$. Then compute $\frac{dL}{d\theta}$ and evaluate $\frac{dL}{d\theta} |_{\theta=0}$. (2 marks)
- (b) Using gradient descent with update $\theta^{(k+1)} = \theta^{(k)} - \eta \frac{dL}{d\theta} |_{\theta=\theta^{(k)}}$, starting from $\theta^{(0)} = 0$:
 - i) Compute $\theta^{(1)}, \theta^{(2)}, \theta^{(3)}$ for learning rate $\eta = 0.1$, (2 marks)
 - ii) Repeat only for $\theta^{(1)}$ with $\eta = 0.5$. (1 marks)
 - iii) Which step size ($\eta = 0.1$ or $\eta = 0.5$) would you choose for this problem? Justify your choice in few sentences (consider stability and speed). (2 marks)
- (c) Solve $\frac{dL}{d\theta} = 0$ to find the exact minimizer θ^* and compute $L(\theta^*)$. Evaluate whether more gradient-descent iterations are warranted here, given you can compute θ^* exactly. (3 marks)

Question 03: Conditional Probability**(10 Marks)**

Email spam filtering is a classic AI task where we update our belief about the class (Spam S vs NotSpam $\neg S$) after observing simple content features. In this question we use one binary feature E : the email contains at least one PromoWord (Yes/No).

You are given a labeled email collection of 100 emails to use in all parts.

- Counts: $|E| = 40 \Rightarrow P(E) = 0.40, P(\neg E) = 0.60$.
- Class priors: $P(S) = 0.30, P(\neg S) = 0.70$.
- Class-conditional rates: $P(E | S) = \frac{24}{30} = 0.80, P(E | \neg S) = \frac{16}{70} \approx 0.2286$.
- Email-collection random variable (PromoWord count): $X \in \{0,1,2\}$ with $P(X = 0) = 0.60, P(X = 1) = 0.30, P(X = 2) = 0.10$.

(a) Compute from the email collection and verify the stated identity:

- $P(E)$ and $P(\neg E)$. (1 marks)
- $P(E \cap S)$ and $P(E \cap \neg S)$. (1 marks)
- Verify the *law of total probability* for E : $P(E) = P(E | S)P(S) + P(E | \neg S)P(\neg S)$. (1 marks)

(b) A new email is observed with PromoWord = Yes (so E is true). Use Bayes' theorem to update the class belief:

- Write the formula for $P(S | E)$ and compute its numerical value using the email-collection rates. (1 marks)
- Compute $P(S | \neg E)$ similarly. (1 marks)
- In one sentence, state which observation (E or $\neg E$) makes Spam more likely in this email collection, and justify with your numbers. (1 marks)

(c) Treat the PromoWord count X as a discrete random variable ($X \in \{0,1,2\}$) for the same email collection.

- Compute expectation $\mathbb{E}[X]$ and variance $\text{Var}(X)$. (2 marks)
- Now assume the PromoWord count $X \in \{0,1,2\}$ follows a Binomial(2, p) model. Estimate p (i.e., $\hat{p}$) from the email collection as

$$\hat{p} = \frac{\text{total promo words observed}}{\text{total slots}} = \frac{0 \cdot 60 + 1 \cdot 30 + 2 \cdot 10}{2 \times 100}.$$

Using this $\hat{p}$: predict $P(E) = P(X \geq 1) = 1 - (1 - \hat{p})^2$, and compare the model's variance $2\hat{p}(1 - \hat{p})$ with your $\text{Var}(X)$ from (c)(i) in one sentence. (2 marks)

Question 04: Hypothesis Testing and Confidence Intervals**(10 Marks)**

An AI-powered email marketing system is being evaluated after deploying a new subject line, referred to as Variant B. The goal is to determine whether Variant B has improved user engagement. The user engagement is measured using the Click-Through Rate (CTR), defined as the proportion of sent emails that received at least one click. A significance level of $\alpha = 0.05$ is used for all tests. Note that H_0 , H_1 are null hypothesis, alternative hypothesis respectively.

Use the following evaluation summaries throughout the question.

- **CTR benchmark:** $p_0 = 0.12$. Variant B: $n = 1000$ emails sent, $x = 150$ clicks.
- **Reading time (opened emails):** $n = 16$, mean $\bar{t} = 18.2$ s, sample standard deviation $s = 4.0$ s. Target mean $\mu_0 = 17$ s.
- **Topic mix (opened emails):** out of $N = 600$, observed counts (Promo, Updates, Personal) = (260,180,160). The expected mix is (0.40,0.30,0.30).

(a) Checking if CTR improved.

- State H_0 and H_1 to test if Variant B's CTR is higher than $p_0 = 0.12$. (1 marks)
- Compute the Z statistic $z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$ (with proportion $\hat{p} = x/n$) and decide at $\alpha = 0.05$ by comparing to the one-tailed critical value $z_{0.95} = 1.645$. (1 marks)
- Now compute 95% Confidence Interval (CI) for p : $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$, and in one sentence say what this CI adds beyond the yes/no decision. (1 marks)

(b) Checking if readers spend longer on B.

- Test $H_0: \mu = 17$ vs $H_1: \mu > 17$ with $t = \frac{\bar{t} - \mu_0}{s/\sqrt{n}}$. State degrees of freedom (df) and decide at $\alpha = 0.05$ by comparing to the one-tailed critical value $t_{0.95, 15} = 1.753$. (1 marks)
- Compute a 95% CI for μ : $\bar{t} \pm t_{0.975, 15} \frac{s}{\sqrt{n}}$ (where $t_{0.975, 15} = 2.131$). Add one short sentence to interpret it. (1 marks)
- Define a Type-I error and a Type-II error for this setting. (1 marks)

(c) Checking if B changed the topic mix.

- Write H_0 and H_1 for a chi-square goodness-of-fit test against the expected mix (0.40,0.30,0.30). (1 marks)
- Compute the expected counts (from the expected mix given), then $\chi^2 = \sum \frac{(O-E)^2}{E}$. State the df. Note that O , E refer to Observed, Expected counts respectively. (1 marks)
- Decide at $\alpha = 0.05$ by comparing your statistic to $\chi_{0.95, 2}^2 = 5.991$. Give one sentence explaining what that means here. (1 marks)
- State whether the p -value is greater than or less than 0.05. In one sentence, explain what this implies about how well the data fits the expected mix. (1 marks)